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2026	6CS012/HJ1: Artificial Intelligence and Machine Learning	1	Report

[Theory / QnA (Machine Learning concepts)]

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Long Question

Discuss the major challenges faced by a Machine Learning Engineer when deploying ML models in a large-scale e-commerce environment, including data drift, latency, and model degradation. Explain their consequences, possible solutions, trade-offs, and the role of MLOps and LLM-based tools.

Being a Machine Learning Engineer in a large-scale online vendor, implementing machine learning algorithms in production raises a number of practical issues. Among the problems is that of data drift where marketing behaviour and transactions change with time since the markets are dynamic (Luo & Yin, 2021). These changes in user interactions can decrease the accuracy of a model used in e-commerce system, like recommendation engines and fraud detection, unless constantly updated. Studies indicate that it is imperative to constantly monitor and re-train pipelines to ensure that a model performs according to the current data distribution since models have to be able to fit an evolving data distribution (Zarour, et al., 2025). Nonetheless, retraining fairly often raises the cost of computation and system complexity, producing a computational efficiency trade-off with system adaptability.

The other critical problem is latency, especially when using real-time applications such as product recommendations and fraud detection. Latency can give rise to a negative user experience and lower conversion rates. To rectify this, organizations employ optimization strategies like model compression, caching and scalable infrastructure. MLOps cultural practices focus on automation and effective deployment pipelines to minimize delays and enhance responsiveness (Zarour, et al., 2025). But, to favor the speed, less complex models can be used but this compromises the accuracy.

The third challenge includes model degradation and monitoring in which model behavior degrades because of invisible patterns or data changes. Such deterioration can manifest itself in deteriorating accuracy or precision over time. Research emphasizes the need to have constant checks and feedback mechanisms to enforce abnormality presence and reliability in manufacturing systems (Zarour, et al., 2025). In the rapidly evolving e-commerce space, inability to track models may cause losses of money and incorrect recommendation. Although monitoring enhances robustness, it entails extra infrastructure, and engineering. Lastly, effective collaboration and MLOps must be implemented to ensure successful deployment. CI/CD pipelines, continual training, and monitoring systems are examples of automation tools that can implement the efficient management of model lifecycle and minimise operational complexity (Zarour, et al., 2025). Emerging LLM-based tools are also useful in debugging, documentation and workflow automation,

increasing productivity and scaling.

Category 1

Describe two techniques to reduce overfitting (dropout, early stopping, etc.) with real-world examples.

Overfitting is the result of a machine learning model learning the training data excessively well, both capturing noise and irrelevant patterns (among other things). There are two methods commonly used to curb overfitting dropout and early stopping.

A regularization method that is commonly used is the dropout, which does not consider the neurons that have been randomly chosen during the training. This will avoid the overreliance of the model on particular features and enhance generalization. In practice, dropout is used in image classification or recommendation systems in real-world scenarios to make sure that the model is not trained on training images, but works correctly with novel data.

Another useful method which keeps a check with the model performance on a validation set in the training process is called early stopping. Training is halted when validation loss begins to grow, whilst training loss becomes smaller. Studies have revealed that tracking the validation loss curves is a valid method of estimating and avoiding the problem of overfitting during training (Li, et al., 2024). When preventing fraud or predicting systems with a deep learning model, such as in a predictive analytics system, early stopping can be used to locate the optimal point to terminate training before it declines.

Both methods enhance model robustness at the cost of increased tuning time and possible underfitting when overfitting.

Category 2

Compare CNN vs. RNN (input type, architecture, parameter sharing, challenges like vanishing gradients).

Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are popular deep learning models, which are tailored to various data and tasks.

The most common use of CNNs is on spatial data e.g. images. They are composed of pooling and convolutional layers that obtain hierarchical attributes of input data. A major benefit of CNNs is parameter sharing, in which the same filter is used in multiple areas which reduces the number of parameters and makes them more efficient (Jin, et al., 2025). The pooling layers can however contribute to the loss of information.

Conversely, RNNs handle sequential or time-based data like text, speech or time-series data. They have memory internally so that, they are able to save the relationship between the previous and current inputs (Jin, et al., 2025). This enables them to be used in applications such as language modeling and prediction. One of the problems in RNNs is the vanishing gradient, in which the long-term dependencies are very hard to learn. Variations on this issue have seen better versions like LSTM and GRU that make use of gating in order to store valuable information across time (Jin, et al., 2025).

Overall, CNNs are favored in terms of spatial feature extraction whereas RNNs are more sequential data, and they have their own advantages and disadvantages.

References

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